

In this Standard the comma has been adopted as the decimal sign

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**EMB-145/314/390/500/505/550/ERJ-170/190
VISUAL DEFECTS – REWORK AND ACCEPTANCE CRITERIA
FOR EXTERNAL AREAS**

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1. SCOPE

This Standard establishes rework procedures and acceptance criteria for visual defects on external surface of aluminum alclad sheet skin parts before painting.

2. APPLICATION

This Standard is applicable for EMB-145, EMB-314, EMB-390, EMB-500, EMB-505, EMB-550, ERJ-170 and ERJ-190 aircrafts to the following defects:

- A) Dent.
- B) Die Punch Mark.
- C) Scratch.
- D) Waviness on riveting line.

NOTE

This Standard is not applicable to slats, lipskins and all leading edges.

3. APPLICABLE DOCUMENTS

Embraer Standards

MEP 13-110 – Solution for Clad Test / Neutralization
MEP 18-005 – Corrugated Paper Masking Tape.
MP 1.4.063 – Aluminum Shim Stock, Laminated.
NE 07-001 – Liquid Penetrant Inspection.
NE 40-012 – Metallic Parts, Cleaning.
NE 40-134 – Sealing of Aircraft, General Requirements.

Foreign Standards

Military Specification (DoD – USA)

MIL-HDBK-728/2 - Military Handbook - Eddy Current Testing.
MIL-STD-2154 - Inspection, Ultrasonic, Wrought Metals, Process for.

National Aerospace Standard

NAS1097 – Rivet, Solid, 100°, Flush Shear Head.
NAS1149 – Washer Flat.

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4. DEFINITIONS

SKIN QUALITY (RAW MATERIAL)

Aluminum sheet surface condition with good shining and reflectivity, free of defects and with finish appearance better than standard roll condition.

CARTOGRAPHY

Document that registers all aircraft external skin non-conformity.

NOTE

The cartography is mandatory to suppliers and shall be replaced for an applicable non-conformity document at Embraer's sites.

CLAD

A layer of pure aluminum that is applied over one or both surfaces of an aluminum alloy sheet for corrosion protection.

CORE

In clad sheets, the base material to which the surface layer is applied.

DEFECT

Any condition that deviates from the ideal. A defect may or may not be acceptable depending upon the type, severity or extension.

MISSING CLAD

An area where the core material has been exposed to external environmental due clad removal.

MRB (MATERIALS REVIEW BOARD)

Engineering liaison members organized by the Quality Engineering that are required to reviews non-conforming materials or parts and determines dispositions for those materials and parts, as well request the necessary corrective actions.

MRB DELEGATION

MRB authority delegated to partners or major subcontractors. In this Standard the same as MRB.

PLUGGED HOLE

A rivet installed to fulfill a hole done to remove a sharp defect or due a cold work repair.

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WAVINESS ON RIVETING AREA

Compound or periodic waves as shown in Figure 1, resulting from the skin riveting operation in the primary structure.

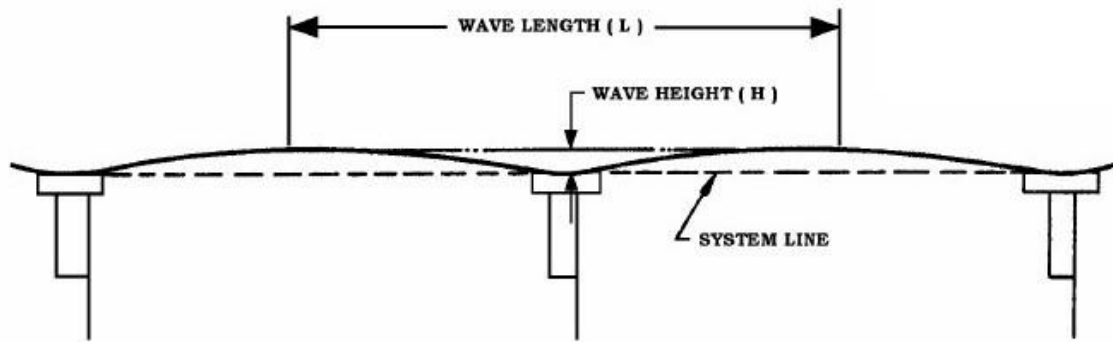


FIGURE 1 : WAVINESS ON RIVETING AREA

4.1 SPECIFIC DEFECTS DEFINITIONS

Dent – An inward or outward defect on the skin, generally caused by impact, without material removal. This impact is separated in two types depending on the kind of the object edge that was responsible for the damage.

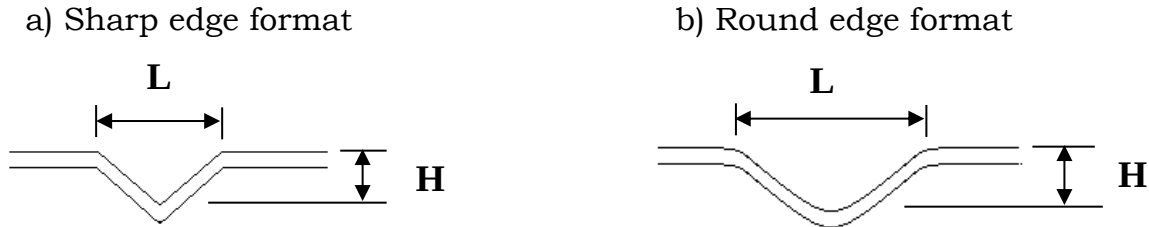


FIGURE 2 : TYPES OF DENT FORMAT

Die punch mark – Inward defect on the skin (from outside to inside), of smooth profile, without material removal, generally caused by die punch hammer impact, around riveting line.

Scratch – A "scratch defect" refers to a visible or tangible imperfection on the surface of an object caused by the scraping, rubbing, or abrading of a harder material against it. It typically results in a shallow, linear mark or groove on the surface.

Skin waviness on riveting area – Waviness on skin with its top and valley among successive rivets or fasteners on a junction area.

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5. CRITERIA AND DECISION TREE

All conditions described herein shall be satisfied, if not, contact the MRB for disposition before any further step.

5.1 AIRCRAFT AREAS DEFINITION

For EMB-145 and ERJ-170 / 190 programs, there are three different acceptance criteria areas according to the visible condition as shown in Figure 3 and described in 5.1.1 to 5.1.6.

For EMB-500 / 505 programs, there are three different acceptance criteria areas according to the visible condition as shown in Figure 4 and described in 5.1.7 to 5.1.9.

For EMB-550 program, there are three different acceptance criteria areas according to the visible condition as shown in Figure 5 and described in 5.1.10 to 5.1.12.

For EMB-390 program, there is only low visibility (LV) acceptance criteria area in the entire aircraft external surface.

For EMB-314 program, there are three different acceptance criteria areas according to the visible condition as shown in Figure 6 and described in 5.1.13 to 5.1.15.

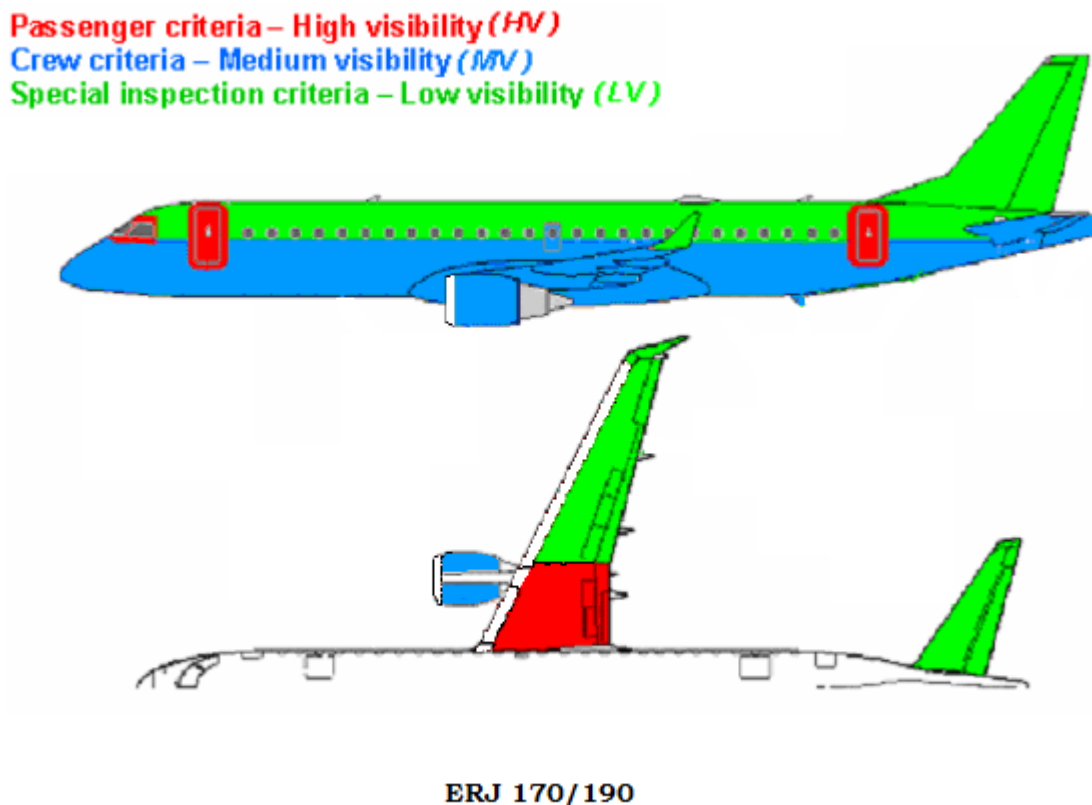


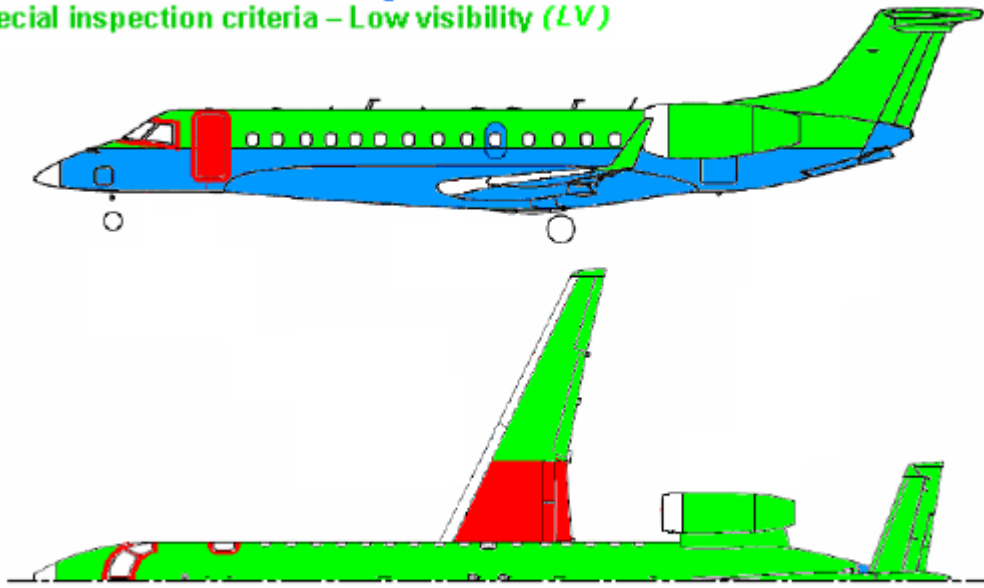
FIGURE 3 : EMB-145 AND ERJ-170/190 AIRCRAFT AREAS DEFINITION

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Passenger criteria – High visibility (HV)
Crew criteria – Medium visibility (MV)
Special inspection criteria – Low visibility (LV)



EMB-145

FIGURE 3: EMB-145 AND ERJ-170/190 AIRCRAFT AREAS DEFINITION

5.1.1 ERJ-170 / 190 High Visibility Area (HV)

On red color, is defined as the most critical one to the visual aspect. This area comprehends the contour of passenger door and service door (800mm to the left and right measuring from the edge of the door and 500mm to the upper side, measured on the same way), contour of windshield (500mm all around) and the wing upper skin from the junction with the fuselage until the 9th rib.

5.1.2 ERJ-170 / 190 Medium Visibility Area (MV)

On blue color, is defined as medium critical to the visual aspect. This area comprehends the all bottom of the fuselage, from 11th LH profile until the 11th RH profile, one side to other, under belly fairings, the entire emergency door, pylon, engine cowl, the wing lower panels, horizontal stabilizers lower skin and its flight controls and fairings.

5.1.3 ERJ-170 / 190 Low Visibility Area (LV)

On green color, is defined as non-critical to the visual aspect. This area comprehends the upper side of the fuselage, vertical stabilizers, the upper skin of horizontal stabilizers and its flight controls and fairings.

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5.1.4 EMB-145 High Visibility Area (HV)

On red color, is defined as the most critical one to the visual aspect. This area comprehends the contour of passenger door and service door (800mm to the left and right measuring from the edge of the door and 500mm to the upper side, measured on the same way), contour of windshield (500mm all around) and the wing upper skin from the junction with the fuselage until the 9th rib.

5.1.5 EMB-145 Medium Visibility Area (MV)

On blue color, is defined as medium critical to the visual aspect. This area comprehends the all bottom of the fuselage, from 10th LH stringer until the 10th RH stringer, one side to other.

The under belly fairings, the entire emergency door, the wing lower panels and its flight controls and fairings.

5.1.6 EMB-145 Low Visibility Area (LV)

On green color, is defined as non-critical to the visual aspect. This area comprehends the upper side of the fuselage, vertical stabilizers and its flight controls, the horizontal stabilizers and its flight controls and fairings, pylon and engine cowl

Passenger criteria – High visibility (HV)
Crew criteria – Medium visibility (MV)
Special inspection criteria – Low visibility (LV)

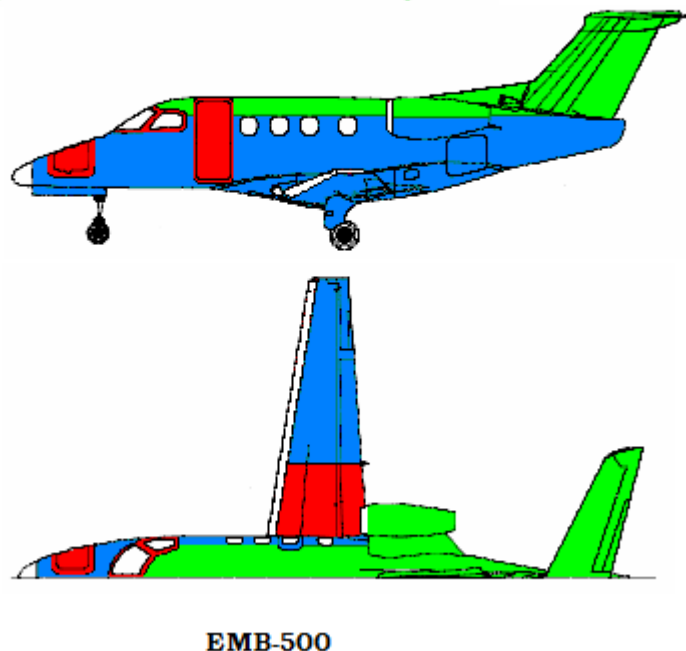


FIGURE 4 : EMB-500 AND EMB-505 AIRCRAFT AREAS DEFINITION

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Passenger criteria – High visibility (HV)
 Crew criteria – Medium visibility (MV)
 Special inspection criteria – Low visibility (LV)

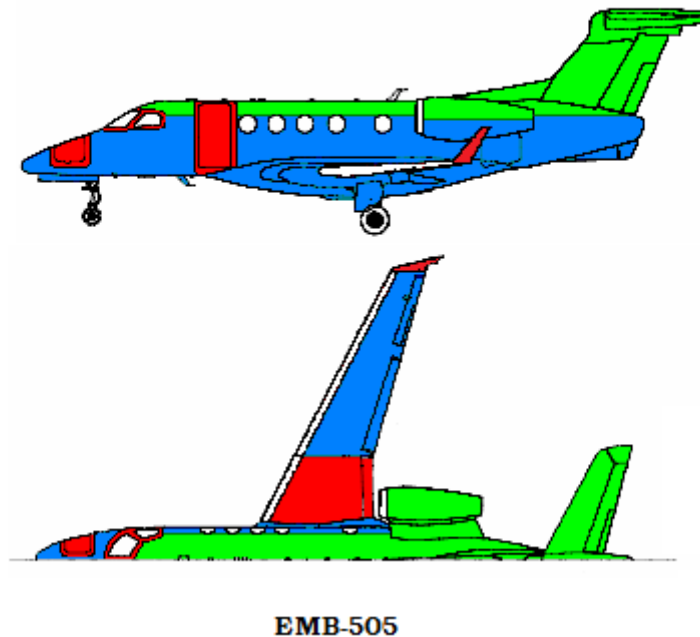


FIGURE 4 : EMB-500 AND EMB-505 AIRCRAFT AREAS DEFINITION

5.1.7 EMB-500 / 505 (PHENOM100/300) High Visibility Area (HV)

On red color, is defined as the most critical one to the visual aspect. This area comprehends the contour of passenger door and service door (800mm to the left and right measuring from the edge of the door and 500mm to the upper side, measured on the same way), contour of windshield (500mm all around), the wing upper skin from the junction with the fuselage until the first fence for the EMB-500 or until the end of the inner flap for the EMB-505 and winglet for the EMB-505.

5.1.8 EMB-500 / 505 (PHENOM100/300) Medium Visibility Area (MV)

On blue color, is defined as medium critical to the visual aspect. This area comprehends the all bottom of the fuselage up to the upper edge of the windows in both fuselage sides, under belly fairings, the entire emergency door, pylon bottom side, engine cowl bottom side, the wing lower panels, wing upper panel from the first fence up to the end of the wing for the EMB-500 or wing upper panel from the end of the inner flap up to the winglet for the EMB-505 and its flight controls.

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5.1.9 EMB-500 / 505 (PHENOM100/300) Low Visibility Area (LV)

On green color, is defined as non-critical to the visual aspect. This area comprehends the upper side of the fuselage, vertical stabilizers and its flight controls, the horizontal stabilizers and its flight controls and fairings, pylon upper side and engine cowl upper side.

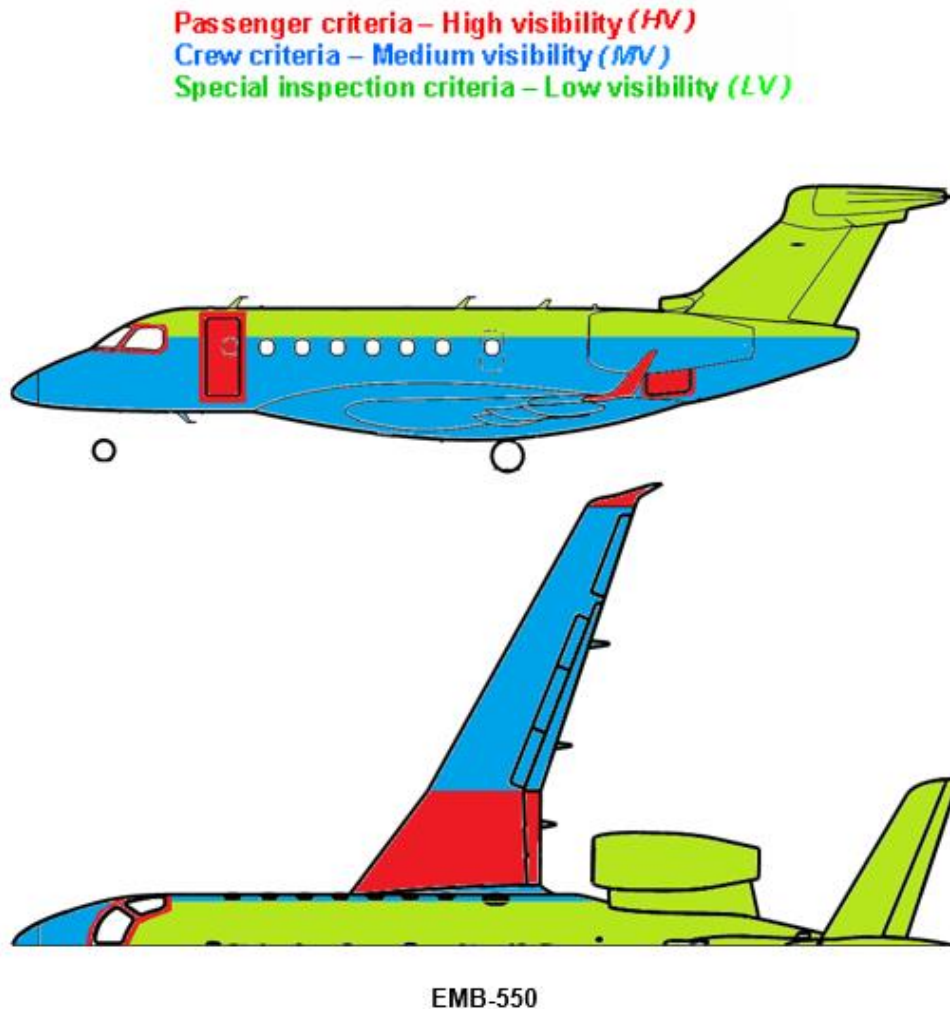


FIGURE 5 : EMB-550 AIRCRAFT AREAS DEFINITION

5.1.10 EMB-550 (LEGACY 450/500) High Visibility Area (HV)

On red color, is defined as the most critical one to the visual aspect. This area comprehends the contour of passenger door and service door (800mm to the left and right measuring from the edge of the door and 500mm to the upper side, measured on the same way), contour of windshield (500mm all around), the wing upper skin from the junction with the fuselage until the end of the inner flap and winglet.

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5.1.11 EMB-550 (LEGACY 450/500) Medium Visibility Area (MV)

On blue color, is defined as medium critical to the visual aspect. This area comprehends the all bottom of the fuselage up to the upper edge of the windows in both fuselage sides, under bally fairings, the entire emergency door, pylon bottom side, engine cowl bottom side, the wing lower panels, wing upper panel from the end of the inner flap up to the winglet and its flight controls.

5.1.12 EMB-550 (LEGACY 450/500) Low Visibility Area (LV)

On green color, is defined as non-critical to the visual aspect. This area comprehends the upper side of the fuselage, vertical stabilizers and its flight controls, the horizontal stabilizers and its flight controls and fairings, pylon upper side and engine cowl upper side.

Ground Criteria – High Visibility (HV)
Crew Criteria – Medium Visibility (MV)
Special Inspection Criteria – Low Visibility (LV)

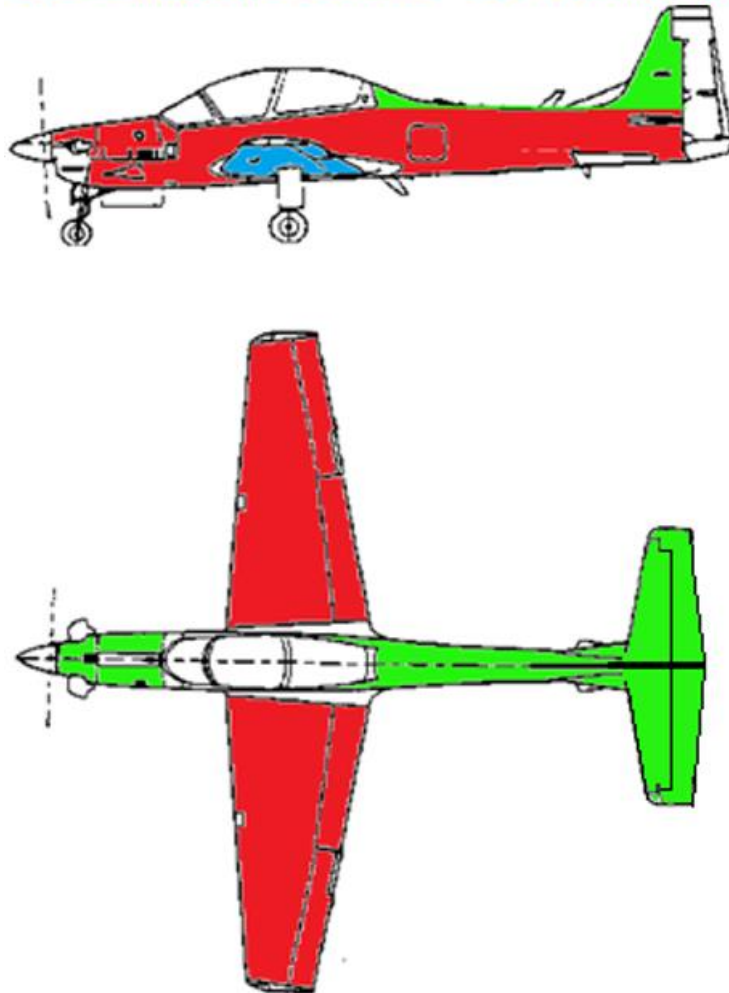


FIGURE 6 : EMB-314 AIRCRAFT AREAS DEFINITION

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5.1.13 EMB-314 (SUPER TUCANO) High Visibility Area (HV)

On red color, is defined as the most critical one to the visual aspect. This area comprehends the wing upper skin and all bottom of the fuselage from stringer 2 LH until the stringer 2 RH.

5.1.14 EMB-314 (SUPER TUCANO) Medium Visibility Area (MV)

On blue color, is defined as medium critical to the visual aspect. This area comprehends the wing lower skin panels, the lower skin of horizontal stabilizers and elevators.

5.1.15 EMB-314 (SUPER TUCANO) Low Visibility Area (LV)

On green color, is defined as non-critical to the visual aspect. This area comprehends the upper side of the fuselage, vertical stabilizers, the upper skin of horizontal stabilizers and elevators.

5.2 PROCEDURE AND ACCEPTANCE CRITERIA ACCORDING TO SEVERITY CLASSIFICATION

The defect herein shall be separated one from other at least of 200mm radio circle, when measured in a way that all the closest defect are enclosed on this defined circle.

5.2.1 Dent

For dents with a sharp edge format or with the relation $H/L > 0,04$ (see Figure 2) the MRB shall give a disposition.

For dents with the relation $H/L \leq 0,04$, follow the Table 1 instructions.

TABLE 1 : DENT ACCEPTANCE CRITERIA AND PROCEDURE INSTRUCTIONS

Aircraft Area	Severity		
	Low	Medium (1)	High
	$H \leq 0,1\text{mm}$	$0,1\text{mm} < H \leq 0,5\text{mm}$	$H > 0,5\text{mm}$
HV and MV	Accept as is	Rework/Repair according to Item 6.2.1.2 and include item on cartography (1)	MRB and include item on cartography (1)
LV	Accept as is	Rework/Repair according to Item 6.2.1.2 and include item on cartography (1)	MRB and include item on cartography (1)

(1) Cartography inclusion only in case of deviation of the requirements specified on drawing.

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5.2.2 Scratch

For scratches, follow the Table 2 instructions.

TABLE 2 : SCRATCH PROCEDURE INSTRUCTIONS (1)

Aircraft Area	Severity	
	Clad not perforated	Clad perforated
HV	Rework/Repair with 1:200 smoothness according to Item 6.2.2.1	MRB and include on cartography
MV and LV	Rework/Repair with 1:100 smoothness according to Item 6.2.2.1	MRB and include on cartography

(1) See Figure 9 for 1:100 or 1:200 smoothness rework/repair instructions.

5.2.3 Die Punch Mark

For die punch marks, follow the Table 3 instructions.

Where : H= mark depth

T_{min} = Minimum sheet thickness according to drawing or applicable standard.

TABLE 3 : DIE PUNCH MARK PROCEDURE INSTRUCTIONS (1)

Aircraft Area	Severity		
	Low	Medium	High
	$H \leq 0,04mm$	$0,04mm < H \leq 0,05T_{min}$	$H > 0,05T_{min}$
HV	Rework/Repair according to Item 6.2.3 and 1:200 smoothness and include on cartography (2)	Rework/Repair with 1:200 smoothness according to Item 6.2.3 and include on cartography. Ensure the final thickness according to drawing tolerance or applicable standard. If the final thickness becomes thinner than specified, send item to MRB	MRB and include on cartography
MV and LV	Accept as is	Rework/Repair with 1:200 smoothness according to Item 6.2.3. Ensure the final thickness according to drawing tolerance or applicable standard. If the final thickness becomes thinner than specified, send item to MRB	MRB and include on cartography

(1) See Figure 9 for 1:100 or 1:200 smoothness rework/repair instructions.

(2) Cartography inclusion, only if the reworked area becomes bigger than 10mm diameter.

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5.2.4 Waviness On Riveting Area

For waviness with the relation $H/L > 0,017$ (see Figure 1), the MRB shall give a disposition.

For waviness with the relation $H/L \leq 0,017$, follow the Table 4 instructions.

TABLE 4 : WAVINESS ACCEPTANCE CRITERIA AND PROCEDURE INSTRUCTIONS

	Severity		
	Low	Medium	High
Aircraft Area	$H \leq 0,2\text{mm}$	$0,2\text{mm} < H \leq 0,3\text{mm}$	$H > 0,3\text{mm}$
HV	Accept as is	Rework/Repair according to Item 6.2.4 in order to reduce the discrepancy to $H < 0,2\text{mm}$.	Rework/Repair according to Item 6.2.4 in order to reduce the discrepancy to $H < 0,2\text{mm}$.
MV and LV	Accept as is	Accept as is and include on cartography	Rework/Repair according to Item 6.2.4 in order to reduce the discrepancy to $H < 0,2\text{mm}$.

5.3 DECISION TREE FOR DEFECTS

The Tables 5 through 8 show the decision tree according to the aircraft area and severity for dents, scratches, die punch marks and waviness on riveting area respectively.

TABLE 5 : DECISION TREE FOR DENTS

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
HV	High	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Medium	Rework according to Item 6.2.1	Verify cracks and buckling	- Item 6.2 requisites accomplished - No crack or buckling is allowed	Close document and include on cartography	Submit to MRB
	Low	Accept as is	-	-	-	-

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TABLE 5 : DECISION TREE FOR DENTS

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
MV	High	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Medium	Rework according to Item 6.2.1	Verify cracks and buckling	Item 6.2 requisites accomplished. No crack or buckling is allowed	Close document and include on cartography	Submit to MRB
	Low	Accept as is	-	-	-	-
LV	High	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Medium	Rework according to Item 6.2.1	Verify cracks and buckling	Item 6.2 requisites accomplished. No crack or buckling is allowed	Close document and include on cartography. Just in case of material removal during repair	Submit to MRB
	Low	Accept as is	-	-	-	-

TABLE 6 : DECISION TREE FOR SCRATCHES

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
HV	High (Clad Perforated)	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Low (Clad Not Perforated)	Rework according to Item 6.2.2 keeping smoothness ratio within 1:200	Missing Clad test	Item 6.2 requisites accomplished. Clad layer shall not be perforated	Close internal document	Submit to MRB

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TABLE 6 : DECISION TREE FOR SCRATCHES

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
MV	High (Clad Perforated)	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Low (Clad Not Perforated)	Rework according to Item 6.2.2 keeping smoothness ratio within 1:100	Missing Clad test	Item 6.2 requisites accomplished. Clad layer shall not be perforated	Close internal document	Submit to MRB
LV	High (Clad Perforated)	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Low (Clad Not Perforated)	Rework according to Item 6.2.2 keeping smoothness ratio within 1:100	Missing Clad test	Item 6.2 requisites accomplished. Clad layer shall not be perforated	Close internal document	Submit to MRB

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TABLE 7 : DECISION TREE FOR DIE PUNCH MARKS

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
HV	High	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Medium	Rework according to Item 6.2.3; keep smoothness ratio within 1:200	Skin thickness, crack, buckling and missing clad test	Item 6.2 requisites accomplished. Buckling is forbidden, skin thickness reduction shall not be below drawing tolerance or applicable standard and clad shall not be perforated	Close document and include on cartography	Submit to MRB
	Low	Rework according to Item 6.2.3; keep smoothness ratio within 1:200	Missing Clad test	Clad layer shall not be perforated	Close document. Include on cartography only if final reworked area becomes bigger than 10mm diameter	Submit to MRB
MV	High	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Medium	Rework according to Item 6.2.3; keep smoothness ratio within 1:200	Skin thickness, crack, buckling and missing clad test	Item 6.2 requisites accomplished. Buckling is forbidden, skin thickness reduction shall not be below drawing tolerance or applicable standard and clad shall not be perforated	Close document and include on cartography	Submit to MRB
	Low	Accept as is	Visual inspection	If visible according to visual aspect criteria, include on cartography	Close document and include on cartography, if necessary	-

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TABLE 7 : DECISION TREE FOR DIE PUNCH MARKS

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
LV	High	MRB	According to MRB request	According to MRB decision	Close document and include on cartography	Submit to MRB
	Medium	Rework according to Item 6.2.3; keep smoothness ratio within 1:100	Skin thickness, crack, buckling and missing clad test	Item 6.2 requisites accomplished. Buckling is forbidden, skin thickness reduction shall not be below drawing tolerance or applicable standard and clad shall not be perforated	Close document and include on cartography	Submit to MRB
	Low	Accept as is	-	-	-	-

TABLE 8 : DECISION TREE FOR WAVINESS

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
HV	High	Rework according to Item 6.2.4, reducing the discrepancy to $H < 0,2\text{mm}$	Measure of H	The waviness shall be eliminated or reduced to limits $H < 0,2\text{mm}$	Close document	Include item on cartography
	Medium	Rework according to Item 6.2.4, reducing the discrepancy to $H < 0,2\text{mm}$	Measure of H	The waviness shall be eliminated or reduced to limits $H < 0,2\text{mm}$	-	Include item on cartography
	Low	Accept as is	-	-	-	-

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continued

TABLE 8 : DECISION TREE FOR WAVINESS

Aircraft Area	Severity	Rework or Repair Procedure	Inspection	Final Condition	Decision if Final Condition is OK	Decision if Final Condition is not OK
MV	High	Rework according to Item 6.2.4, reducing the discrepancy to $H < 0,2\text{mm}$	Measure of H	The waviness shall be eliminated or reduced to limits $H < 0,2\text{mm}$	Close document	Include item on cartography
	Medium	Accept as is	-	-	Include item on cartography	-
	Low	Accept as is	-	-	-	-
LV	High	Rework according to Item 6.2.4, reducing the discrepancy to $H < 0,2\text{mm}$	Measure of H	The waviness shall be eliminated or reduced to limits $H < 0,2\text{mm}$	Close document	Include item on cartography
	Medium	Accept as is	-	-	Include item on cartography	-
	Low	Accept as is	-	-	-	-

6. PARTS REPAIR / REWORK

6.1 MISSING CLAD TEST

To determine if a scratch, score or impact has or has not perforated the clad layer, the following test procedure shall be performed on the affected area:

- solvent clean the surface according to NE 40-012. Before solvent evaporation from the surface, dry it with another clean and dry cotton cloth;
- dry the surface and mask with a tape (MEP 18-005) around the defect to a maximum distance of 3mm from the center of the scratch;
- tear down at the damaged area, one drop of the reactive solution described below:

reactive solution:

200 grams of potassium nitrate,
+ 100 grams of sodium hydroxide,
+ distilled water to complete 1 liter of solution,
(application time: 1 to 3 minutes);

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NOTE

The reactive solution Type I according to MEP 13-110 may be used as alternative of the reactive solution prepared above.

d) allow the solution to remain on the area for the corresponding time period specified above;

e) rinse the surface with clean water (without rubbing action) and observe the affected area:

- a scratch that perforated the clad will be indicated by the exposed core,
- if the part material goes turning dark, the clad was perforated;
- if the part material shows no discoloration, the clad was not perforated;

f) eliminate the dark deposit in the bottom of the scratch (only when the clad layer is perforated) with one or two drops of a nitric acid solution which consist of 1 volume of nitric acid + (2-3) volumes of distilled water (neutralization solution) and wait for half or one minute before rinsing;

NOTE

The neutralization solution Type II according to MEP 13-110 may be used as alternative of the neutralization solution prepared above.

g) carefully rinse and dry the surface (in order to prevent corrosion).

6.2 SKIN DEFECT REWORK AND REPAIR PROCEDURE

All conditions described herein shall be satisfied, if not, contact the MRB for disposition before any further step.

6.2.1 Dent

Perform a visual inspection to verify the type of dent format (see Figure 2). For sharp edge format, rework the affected area according to Item 6.2.1.1 and for round edge format the rework shall be according to Item 6.2.1.2.

6.2.1.1 Dent with sharp edge format for single parts

- a) from the dent position, check edge distance. The edge distance shall be bigger than 2 times the diameter of the closest hole on the skin;
- b) from the dent position, check pitch distance. The pitch shall be bigger than 3 times the diameter of the closest hole on the skin;
- c) the skin thickness shall be at least 0,8 mm;
- d) drill a 3/32" hole centered at the mark;

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e) after performing the hole, proceed according to a localized cold work procedure as described in 6.3 of this Standard.

f) install a 3/32" MS14218AD Brilles rivet or wet install NAS1097 rivet with sealant PS870C12 according to NE 40-134. If the reworked area has a thickness less than 1mm, install the rivet with a suitable internal washer (NAS1149D);

NOTE

If the skin has a thickness of 1,3mm or more after the localized cold work, a 1/8" rivet may be used if necessary, in order to remove any mark that remains;

g) apply sealant PR1776B2 according to NE 40-134 over the rivet shop head (at the inner side of the skin).

ATTENTION

Only one plugged hole is allowed on a square area limited by two stringers and two frames on sequence, for example the square defined by stringer 4LH; 5LH and frames 46; 47.

6.2.1.2 Dent with round edge format

a) perform the cold work procedure described in 6.3;

b) after cold work conclusion, inspect by eddy current according to MIL-HDBK-728 for crack detection and perform missing clad test according to Item 6.1 in the affected area;

c) if any crack or missing clad area was detected, contact the MRB for disposition before any further step;

d) if no crack and no missing clad area was detected, apply surface protection in the reworked area according to specified on drawing.

6.2.2 Scratch

Perform the missing clad test according to 6.1 to define rework according to Item 6.2.2.1 for scratch that does not perforate the clad layer, or rework according to Item 6.2.2.2 for scratch that perforates the clad layer.

6.2.2.1 Scratch that does not perforate the clad layer.

a) solvent clean the damaged area according to NE 40-012, using a clean cloth and wipe gently to avoid scratching;

b)The rework shall be performed according to one of the four methods described as follows:

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ATTENTION

DO NOT USE METALLIC TOOLS LIKE BRUSHES, METAL SCRAPERS, CHIPCHASERS, PICKS, SCREWDRIVERS, BLADES, STEEL WOOL OR SEVERE ABRASIVES TO REMOVE BURRS AND PILE UPS.

THESE TOOLS MAY CAUSE SCRATCHES OR CRACKS AND AS A RESULT, SERIOUS STRUCTURAL DAMAGE TO THE AIRCRAFT MAY OCCUR.

Method 1:

Use of abrasive means to remove material from surface, up to reach the bottom of the scratch (see Figure 10).

- sand the affected area using aluminum oxide sandpaper 320 up to 600 grit until the defect has been completely eliminated;
- after sanding, visual inspect the reworked area with an 8x lens to ensure the defect elimination.
- perform the missing clad test to certify that the clad layer is not trespassed.



FIGURE 7 : SCRATCH REWORKED BY METHOD 1

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6.2.2.2 Scratch that perforate the clad layer (rework only after MRB instruction)

- a) sand the affected area using aluminum oxide sandpaper 320 up to 600 grit until the defect has been completely eliminated;
- b) after sanding, visually inspect the reworked area with an 8x lens to ensure the defect elimination;
- c) perform ultrasonic inspection according to MIL-STD-2154 to measure the skin thickness in the reworked area. If the measured thickness is less than the minimum thickness specified on drawing or in the applicable standard, contact the MRB for disposition before any further step, if not;
- d) protect the reworked area according to specified on drawing.

6.2.3 Die Punch Mark

- a) measure the height “H” of the defect (see Item 5.2.3 and Figure 8);
- b) the area to be reworked shall be determined considering the smoothness criteria of Table 3 for the “H” measured on previous item “a” and passing the circle area obtained from the beginning to the end of the mark line path (see Figure 8);
- c) remove all the fasteners that are enclosed on the area that will be reworked;
- d) sand the area using aluminum oxide sandpaper 320 up to 600 grit until the defect is completely eliminated.
- e) inspect by eddy current according to MIL-HDBK-728 for crack detection and perform missing clad test according to Item 6.1;
- f) if the clad layer is affected, measure the skin thickness and contact the MRB for disposition before any further step, if not;
- g) Install again all the removed fasteners according to specified on drawing; and
- h) apply surface protection in the affected area according to specified on drawing.

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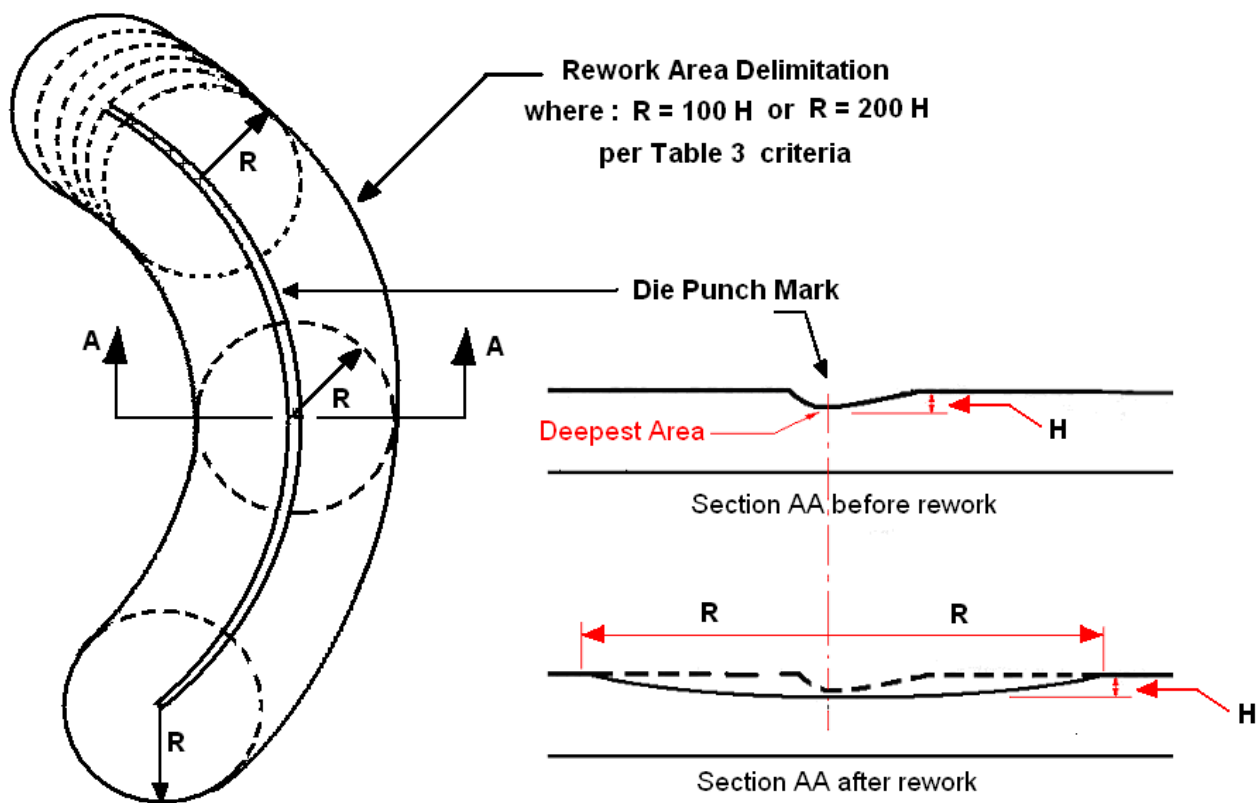
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NOTE

Perform smooth transition between reworked and non-reworked area.

FIGURE 8 : DIE PUNCH MARK REWORK AREA DELIMITATION

6.2.4 Waviness on Riveting Line

- remove the fasteners in the waviness occurrence area, for a maximum of 10 fasteners on the same sequence line;
 - measure the gap between the frame or stringer and the skin (if there was a previous shim installed, it has to be removed before measuring the gap);
- EMB-145, EMB-314, EMB-390, EMB-500, EMB-505, ERJ-170 and ERJ-190 programs: the acceptable gap is up to 0,8mm. For greater values, contact the MRB for disposition before any further step.
 - EMB-550 program: the acceptable gap shall be according to Table 9. The maximum gap is related to the fastener diameter. For greater values, contact the MRB for disposition before any further step.

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TABLE 9 : MAXIMUM GAP FOR THE EMB-550

Fastener Diameter (in)	Maximum Acceptable Gap (mm)
1/8	0,4
5/32	0,5
3/16	0,6
1/4	0,8

c) install a laminated shim type 1 according to MP 1.4.063 with interface sealant PS870C according to NE 40-134. The requirements of peel able shim maximum thickness shall be according to Item 6.2.4 (b);

d) Reinstall the removed fasteners according to drawing.

6.3 LOCALIZED COLD WORK PROCEDURE

6.3.1 Equipment

Figures 9 and 10 show the equipment for localized cold work, consist of:

- Wood bar covered by a 2,0mm rubber layer;
- Pair of small magnets;
- Wood hammer; and
- Tape (metallic speed tape or similar).

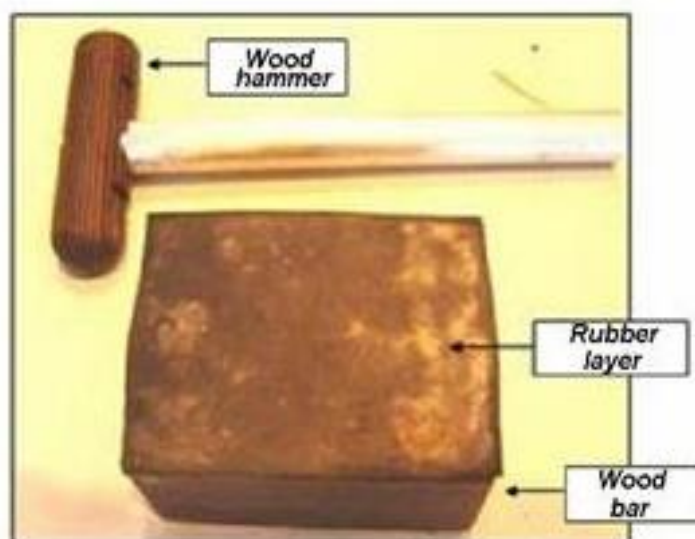


FIGURE 9 : EQUIPMENT FOR LOCALIZED COLD WORK

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6.3.2 Operacional Procedure

Perform localized cold work operation (see Figure 10) according to the following steps:

- Localize the exactly position of the damage on both sides of the fuselage skin using a pair of magnets;
- put a tape over the mark at the protuberance side of the sheet and ensure that the higher point was identified over the tape;
- put the backing bar in the opposite side of the sheet at the damage position;
- work as light as possible with the hammer to restore the original system line;
- remove the tape and perform a visual inspection; and
- inspect by eddy current according to MIL-HDBK-728 for crack detection.

NOTE

The wood hammer shall be free of sharp edges, embedded materials or any defect that may cause some damage to the reworked part.



FIGURE 10 : LOCALIZED COLD WORK OPERATION

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7. TESTS AND INSPECTIONS REQUIREMENTS

7.1 GENERAL REQUIREMENTS

All reworked parts shall conform to the requirements of this specification. If materials, equipment, plant or process do not comply with the requirements of this specification, the process shall be discontinued until the requirements are complied again.

Each part shall be inspected as required in the item 7.5. Parts with unacceptable defects shall be rejected or may be submitted to MRB analysis and disposition.

7.2 INSPECTION EQUIPMENT CONTROL

Equipment used for inspection shall be periodically checked, according to the applicable specification, in order to ensure that all requirements are met.

Instrumentation used to measure and control shall be accurate and calibrated according to the applicable specification.

7.3 QUALIFICATION OF THE PERSONNEL

Only qualified personnel may rework or repair damage clad aluminum sheet parts surfaces.

7.3.1 Visual Acuity Examination Requirements

Prior to being qualified, personnel shall meet minimum acceptance standards for visual acuity as follows:

- a) normal vision : Ortho Rater N°2 or equivalent at least one eye, either natural or corrected, as determined by medically qualified personnel;
- b) color conception: Personnel shall be capable of adequately distinguishing and differentiating colors used in the process involved.

7.4 TEST INSPECTION

7.4.1. Visual inspection

Examination of surface discontinuities that may be observed on finished products shall considerer of the following conditions determined below:

- a) structural discontinuities - Examination of surface discontinuities which may affect structural integrity of finished part. It shall be initially done by unaided eye in order to localize the discontinuities. The examination shall be performed under luminosity condition with of 600 lux minimum by an operator, with a several angles and distances as necessary to guarantee surface quality.

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Detailed inspection using a magnification glass shall be completed in order to identify the discontinuity type with length, width and depth. Additional magnification lens may be used in order to determine the extension of discontinuities for definition purpose as corrosion pits or scratches.

Complementary detailed inspection may be performed using adequate inspection equipments or measurements procedures or chemical attack in order to guarantee discontinuities identification.

b) The visual aspects inspection shall be performed by a person that does not have gotten close to the fuselage any time before the time of the visual inspection. The light condition shall be 600 lux minimum, 2m from the skin and on an angle between 30 degrees and 150 degrees measured from the skin system line.

7.4.2 Non destructive clad thickness inspection

The measurement shall be performed by ultrasonic inspection, according to MIL-STD-2154.

7.5 INSPECTION OF SERIAL PRODUCTION PARTS

All serial production parts shall be submitted to inspection as follows:

7.5.1 Scratches

Visual inspection (see item 7.4.1), if necessary use an 8x lens.

Dial Indicator with Needle Point, in contact point to measure at least 1µm.

7.5.2 Residual Clad Thickness Inspection

The measurement shall be performed to ensure the thickness of cladding removed. This shall be carried out by Ultra Sonic inspection in all reworked areas.

7.5.3 Crack Detection Test

Crack detection test shall be performed to assure that the reworked area has no cracks. This shall be carried out by Eddy Current inspection according to standard MIL-HDBK-728/2, or Liquid Penetrant inspection according to NE 07-001.

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